

INPUTS

IoT Telemetry Streams

DT State Updates

IIoT Operational
(HAI, BATADAL, ICS-Flow)

Failure Injection Events

FlexSim Simulation Traces



PROCESSING PIPELINE

1Ingestion + ID
 $id = H(x || t_0)$ [Eq.1]

2ML Privacy Routing
Route: Fabric or Ethereum

3Concurrent Fabric Commit
 $T_{commit} = endorse + order + validate$ [Eq.5]

4Merkle-Batched Anchoring
 $R = \text{MerkleRoot}\{h_i\}$ [Eq.3]

5HADC Delta Compression
UCB1 policy pi_c per class [Eq.14]

6Checkpoint Versioning
 $E[T_{rb}] = T_{verify} + u * T_{delta}$ [Eq.11]



OUTPUTS

Committed Fabric State

Ethereum Audit Anchors

Compressed Delta Logs

Recoverable Version Chain

Exactly-Once Guarantees